PHYSICS — CLASS XII (Sample syllabus for testing; not an official board document)

Unit I: Electrostatics (Marks: 16)

Chapter 1: Electric Charges and Fields — Electric charges; conservation of charge; Coulomb's law; electric field; electric field lines; electric dipole; Gauss's theorem and its applications.

Chapter 2: Electrostatic Potential and Capacitance — Electric potential; potential difference; equipotential surfaces; capacitors and capacitance; combination of capacitors in series and parallel.

Unit II: Current Electricity (Marks: 12)

Chapter 3: Current Electricity — Electric current; drift velocity; Ohm's law; resistivity and its temperature dependence; Kirchhoff's laws; Wheatstone bridge; potentiometer.

Unit III: Magnetic Effects of Current and Magnetism

Chapter 4: Moving Charges and Magnetism — Biot-Savart law; Ampere's circuital law; force on a moving charge in a magnetic field; moving coil galvanometer.

Assessment

Theory examination: 70 marks, 3 hours.

Practical examination: 30 marks (experiments, project and viva).